Chapter 1

Smoke test

1.1 S. I. Gelfand, 1981-11-23: Jantzen forms and category O

record 540; source pages P000996–P000997.

English and русская цитата: Что если сменить направление?

1981-540.1 Known result and exact verification Class B

Statement. For $A \in M_n(\mathbb{C})$, $\|A\|_{\mathrm{op}} \le \|A\|_F$, and $\mathfrak{g} = \mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+$.
Fate. See Williamson (2016, §8.4).

Proposition 1.1 (An elementary identity)**.** [proved here]$\det \left(\begin{smallmatrix} 1 & t \\ t & t^2+t^3 \end{smallmatrix}\right) = t^3$.

Proof. Expand the determinant. □

Direction	Layers	Result
ρ	$(7,3,2,1)$	verified

Bibliography

Geordie Williamson. Local hodge theory of soergel bimodules. *Acta Mathematica*, 217 (2):341–404, 2016. doi: 10.1007/s11511-017-0146-8. URL https://arxiv.org/abs/1410.2028. verified.